

Q1. Tell me about yourself

A. Provide a concise overview of your background, technical skills, key achievements, and career goals relevant to the role.

Q2. Why do you want this role?

A. Explain how the position aligns with your skills, interests, and long-term career objectives while showing enthusiasm for the company.

Q3. Explain OOP principles

A. The four main OOP principles are Encapsulation, Inheritance, Polymorphism, and Abstraction.

Q4. What is polymorphism?

A. Polymorphism allows objects of different classes to be treated through a common interface while exhibiting different behaviors.

Q5. What is inheritance?

A. Inheritance enables a class to acquire properties and methods from another class, promoting code reuse.

Q6. What is encapsulation?

A. Encapsulation bundles data and methods together while restricting direct access to internal implementation details.

Q7. What is abstraction?

A. Abstraction hides complex implementation details and exposes only essential functionality.

Q8. Array vs linked list?

A. Arrays provide fast indexed access but fixed sizing, while linked lists offer dynamic sizing with slower random access.

Q9. Stack vs queue?

A. A stack follows LIFO (Last In, First Out), while a queue follows FIFO (First In, First Out).

Q10. What is a hash table?

A. A hash table stores key-value pairs and provides efficient lookup, insertion, and deletion operations.

Q11. What is Big O notation?

A. Big O notation describes the time or space complexity of an algorithm as input size grows.

Q12. Explain binary search

A. Binary search repeatedly divides a sorted dataset in half until the target value is found.

Q13. Explain merge sort

A. Merge sort is a divide-and-conquer algorithm that recursively splits and merges data with $O(n \log n)$ complexity.

Q14. Explain quicksort

A. Quicksort partitions data around a pivot and recursively sorts subarrays, with average complexity of $O(n \log n)$.

Q15. What is recursion?

A. Recursion is a technique where a function calls itself to solve smaller instances of a problem.

Q16. Process vs thread?

A. A process is an independent program in execution, while a thread is a lightweight unit of execution within a process.

Q17. What is multithreading?

A. Multithreading allows multiple threads to run concurrently within a single process.

Q18. What is a deadlock?

A. A deadlock occurs when two or more processes wait indefinitely for resources held by each other.

Q19. Race condition?

A. A race condition occurs when multiple threads access shared data simultaneously, causing unpredictable results.

Q20. What is synchronization?

A. Synchronization ensures safe access to shared resources in concurrent environments.

Q21. Explain REST APIs

A. REST APIs use HTTP methods and stateless communication to enable interaction between systems.

Q22. PUT vs PATCH?

A. PUT replaces an entire resource, while PATCH updates only specific fields of a resource.

Q23. What is JSON?

A. JSON is a lightweight data-interchange format used for storing and exchanging structured data.

Q24. Explain MVC

A. MVC separates applications into Model, View, and Controller components for better organization and maintainability.

Q25. Dependency injection?

A. Dependency Injection provides required dependencies externally rather than creating them internally.

Q26. Explain SQL joins

A. SQL joins combine data from multiple tables using related columns such as INNER, LEFT, RIGHT, and FULL joins.

Q27. What is normalization?

A. Normalization organizes database data to reduce redundancy and improve integrity.

Q28. SQL vs NoSQL?

A. SQL databases use structured schemas and relationships, while NoSQL databases offer flexible schemas and scalability.

Q29. What is indexing?

A. Indexing improves database query performance by enabling faster data retrieval.

Q30. What is caching?

A. Caching stores frequently accessed data in fast storage to reduce latency and improve performance.

Q31. Git branching?

A. Git branching allows developers to work on features or fixes independently without affecting the main codebase.

Q32. Merge vs rebase?

A. Merge combines branch histories, while rebase rewrites commit history to create a linear timeline.

Q33. What is CI/CD?

A. CI/CD automates code integration, testing, and deployment to improve software delivery speed and reliability.

Q34. Unit testing?

A. Unit testing verifies individual components or functions in isolation.

Q35. Integration testing?

A. Integration testing validates interactions between multiple system components.

Q36. What is TDD?

A. Test-Driven Development is a methodology where tests are written before implementation code.

Q37. Design patterns?

A. Design patterns are reusable solutions to common software design problems.

Q38. Singleton pattern?

A. The Singleton pattern ensures only one instance of a class exists throughout an application.

Q39. Factory pattern?

A. The Factory pattern creates objects without exposing the creation logic to the client.

Q40. Microservices architecture?

A. Microservices architecture divides applications into small, independent services that communicate through APIs.

Q41. Monolith vs microservices?

A. Monolithic applications are built as a single unit, while microservices consist of independent deployable services.

Q42. What is Docker?

A. Docker packages applications and dependencies into portable containers for consistent deployment.

Q43. What is Kubernetes?

A. Kubernetes automates deployment, scaling, and management of containerized applications.

Q44. Load balancing?

A. Load balancing distributes traffic across multiple servers to improve availability and performance.

Q45. Horizontal scaling?

A. Horizontal scaling adds more servers or instances to handle increased demand.

Q46. Vertical scaling?

A. Vertical scaling increases resources such as CPU, memory, or storage on an existing server.

Q47. Describe a challenging bug

A. Explain the issue, debugging approach, root cause, solution, and lessons learned.

Q48. Describe a team conflict

A. Discuss the conflict, communication strategy, resolution process, and positive outcome.

Q49. Describe a successful project

A. Highlight the objective, your role, technical challenges, results, and business impact.

Q50. Questions for us?

A. Ask about team culture, development practices, project challenges, growth opportunities, and success metrics.